



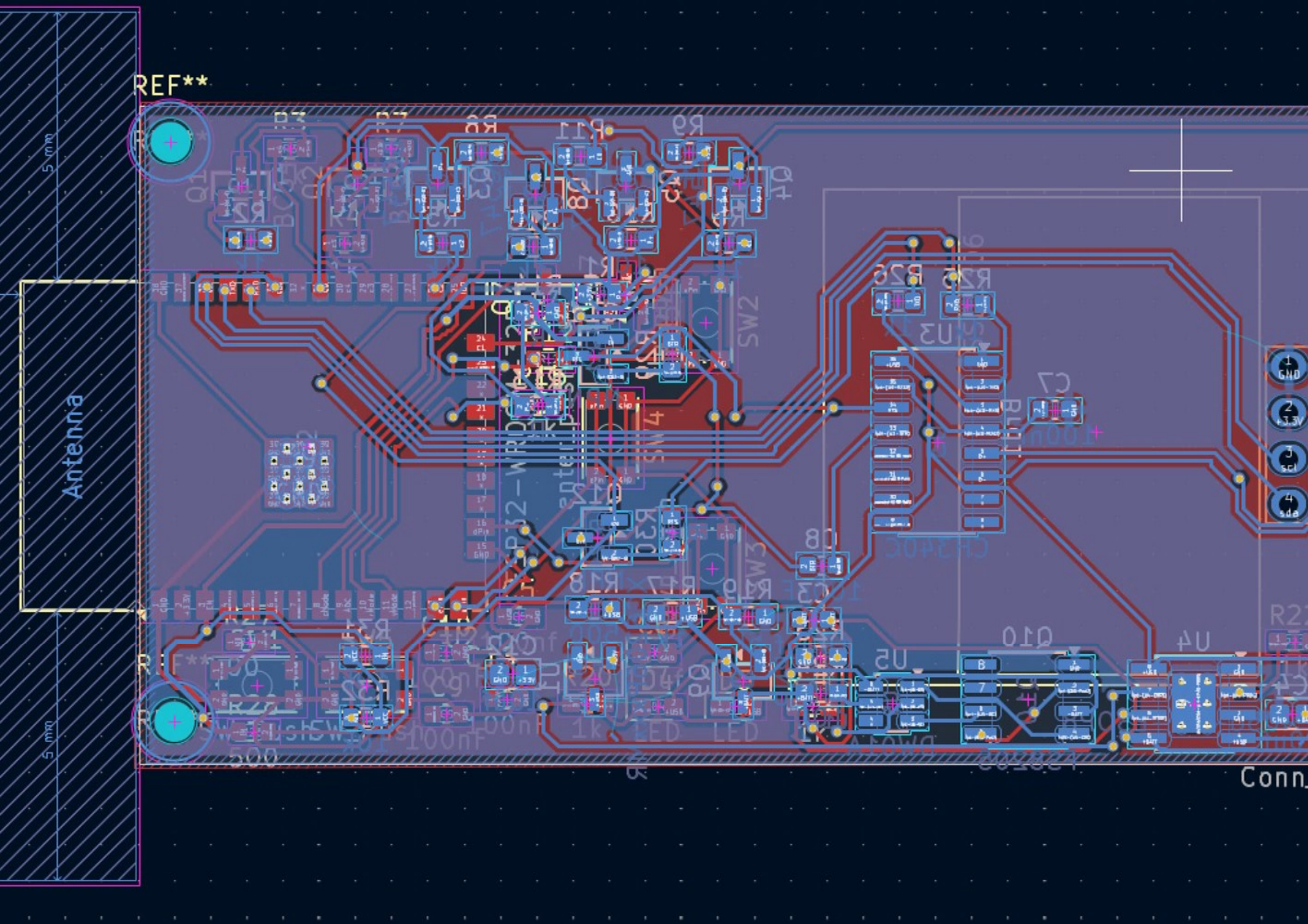
The screenshot shows the Kicad PCB Editor interface. The top toolbar contains icons for various tools: a mouse cursor, a delete icon, a copy icon, a paste icon, a line tool, a arc tool, a rectangle tool, a circle tool, a polygon tool, a text tool, a dimension tool, a footprint tool, a track tool, a zone tool, a pad tool, a dimension tool, and a points tool. The left sidebar shows the 'Layers' panel with a list of layers and their visibility status, and the 'Selection Filter' panel with checkboxes for different object types.

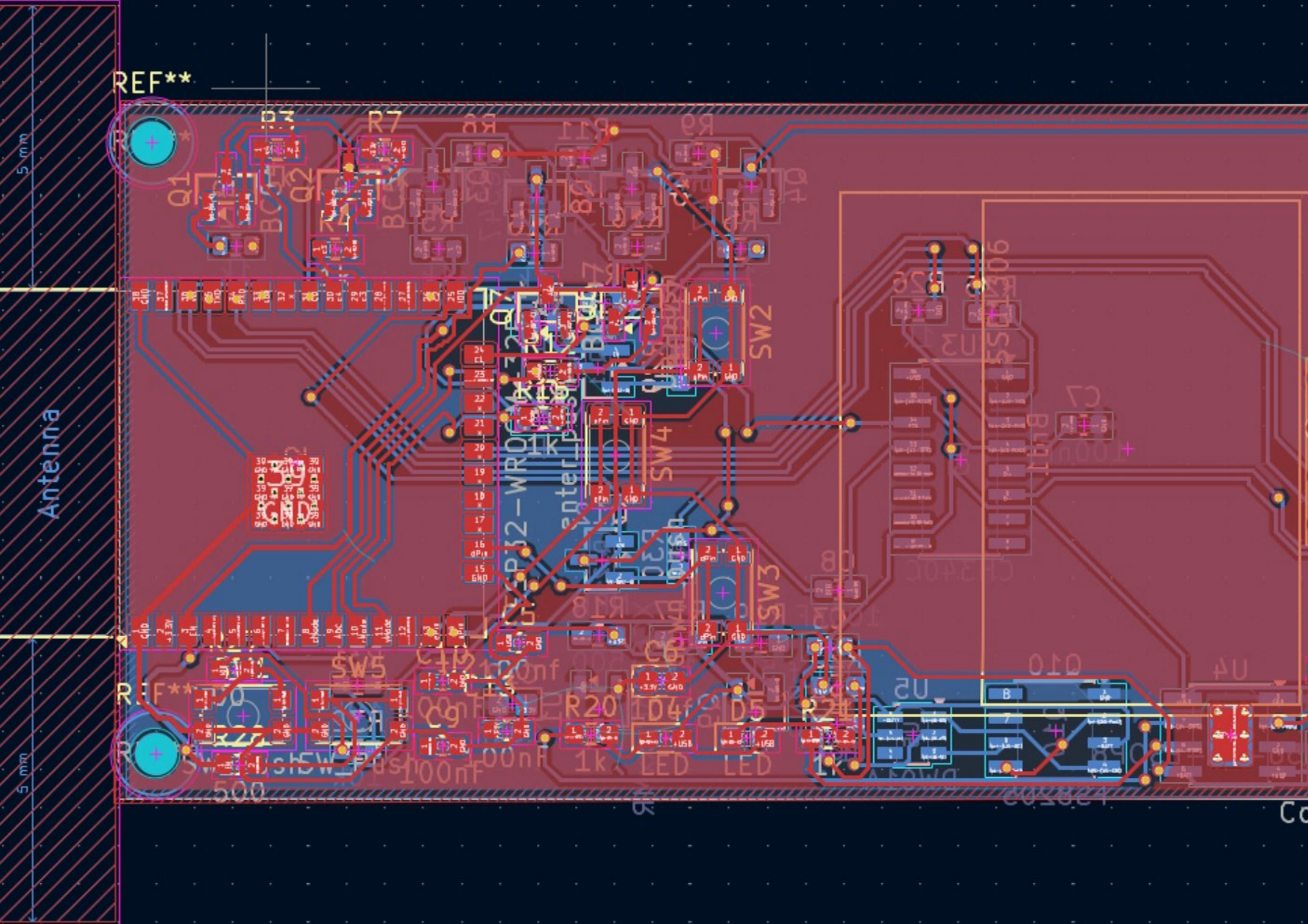
Layers Panel:

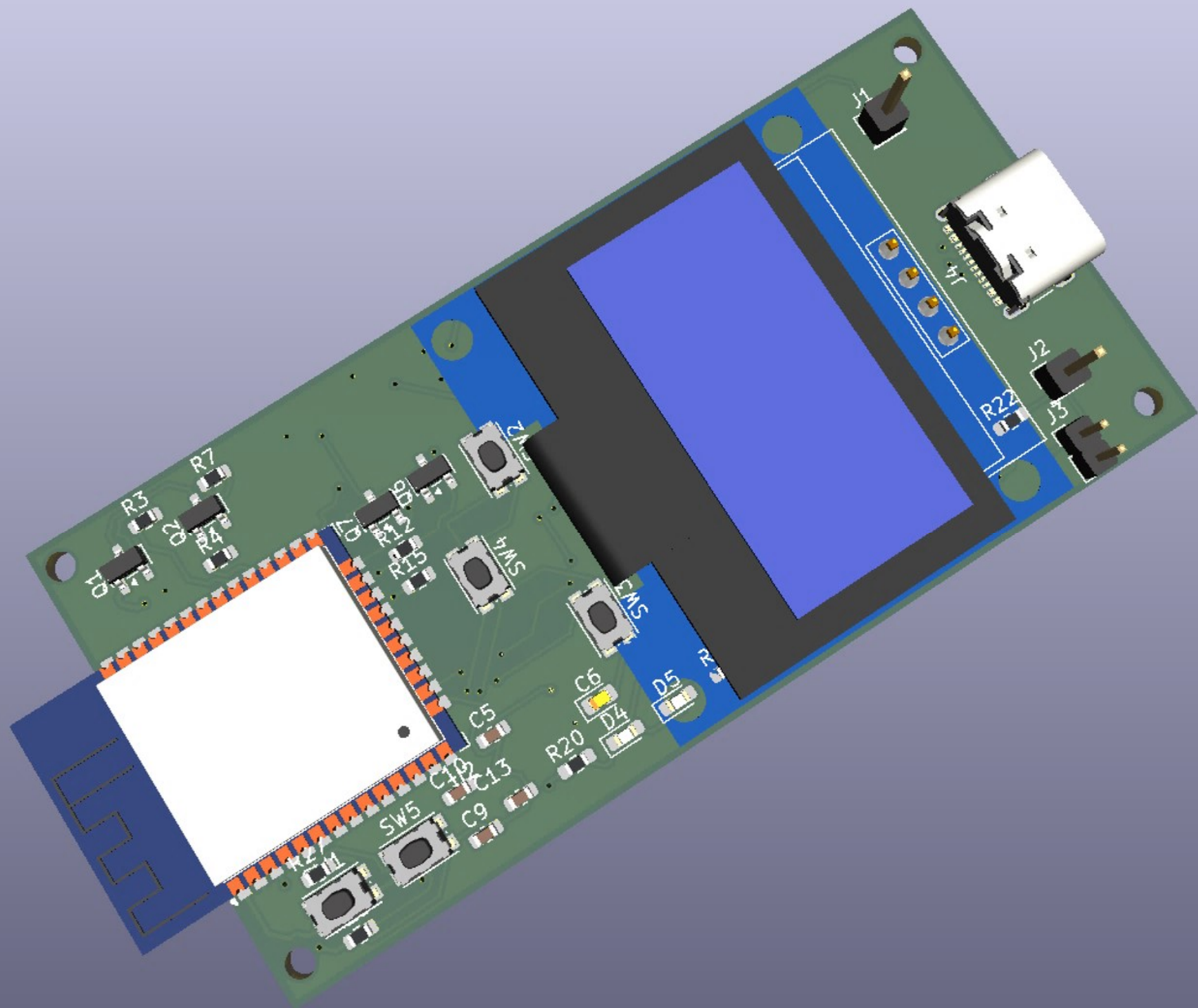
Layer	Visibility
F.Cu	Visible
B.Cu	Visible
F.Ad	Visible
B.Ad	Visible
F.Pas	Visible
B.Pas	Visible
F.Sil	Visible
B.Sil	Visible
F.Ma	Visible
B.Ma	Visible
User	Visible
User	Visible
User	Visible
User	Visible
Edge	Visible
Marg	Visible

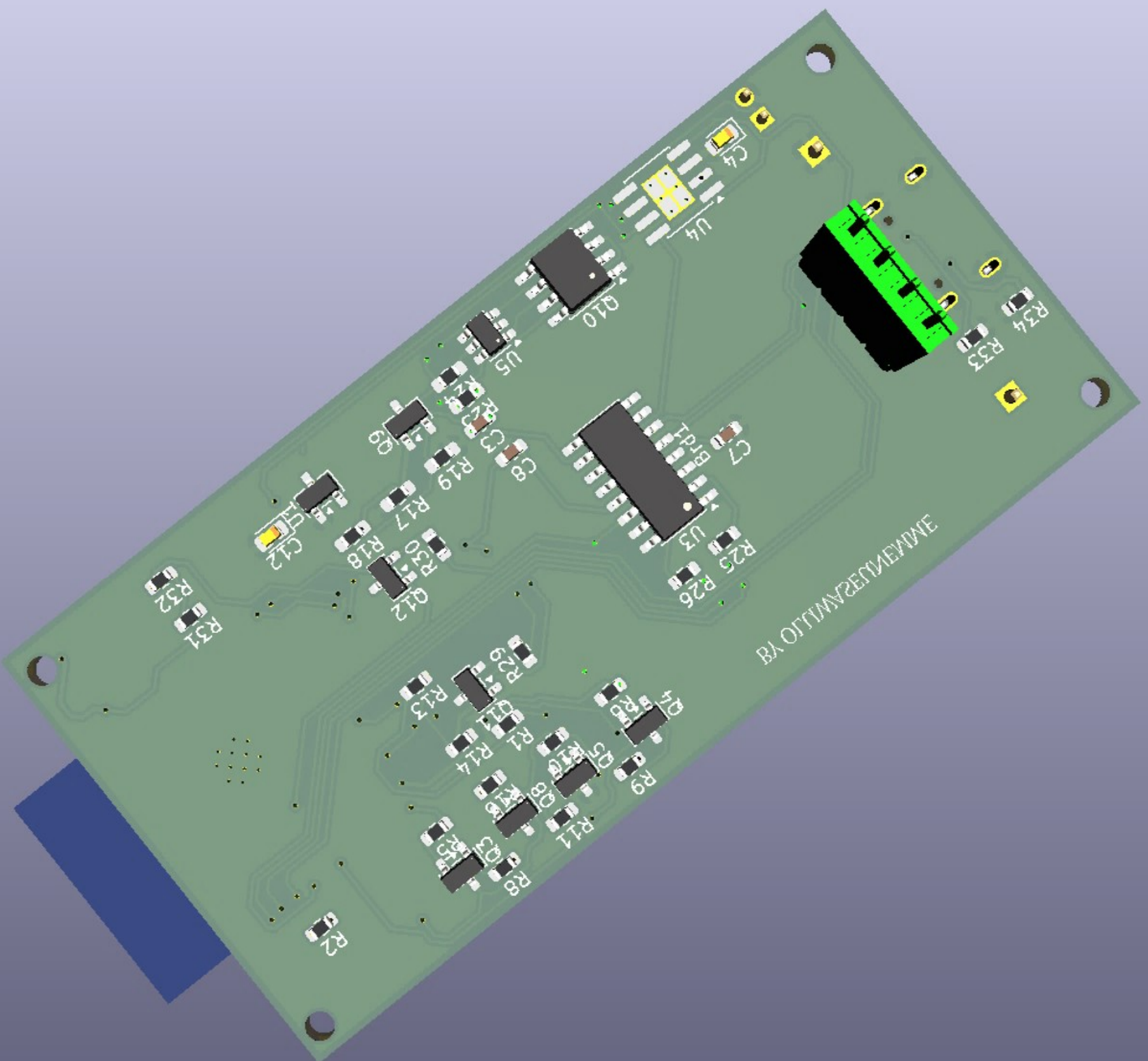
Selection Filter Panel:

- ☒ All items
- ☒ Footprints
- ☒ Tracks
- ☒ Pads
- ☒ Zones
- ☒ Dimension
- ☒ Points









Capacitance

1.02

CALCULATE

RESULT: 1.00nF/1000.0pF,

...

BAND1

BLACK

BROWN

RED

BAND2

BLACK

RED

BAND3

BROWN

RED

ORANGE

MULTIPLIER

BROWN

RED

ORANGE

TOLERANCE

WHITE

GOLD

SILVER

RESULT:

12.00K Ω , $\pm$ 5%

CALCULATE



CONNECTION CALCULATOR

Types:



Series



Parallel

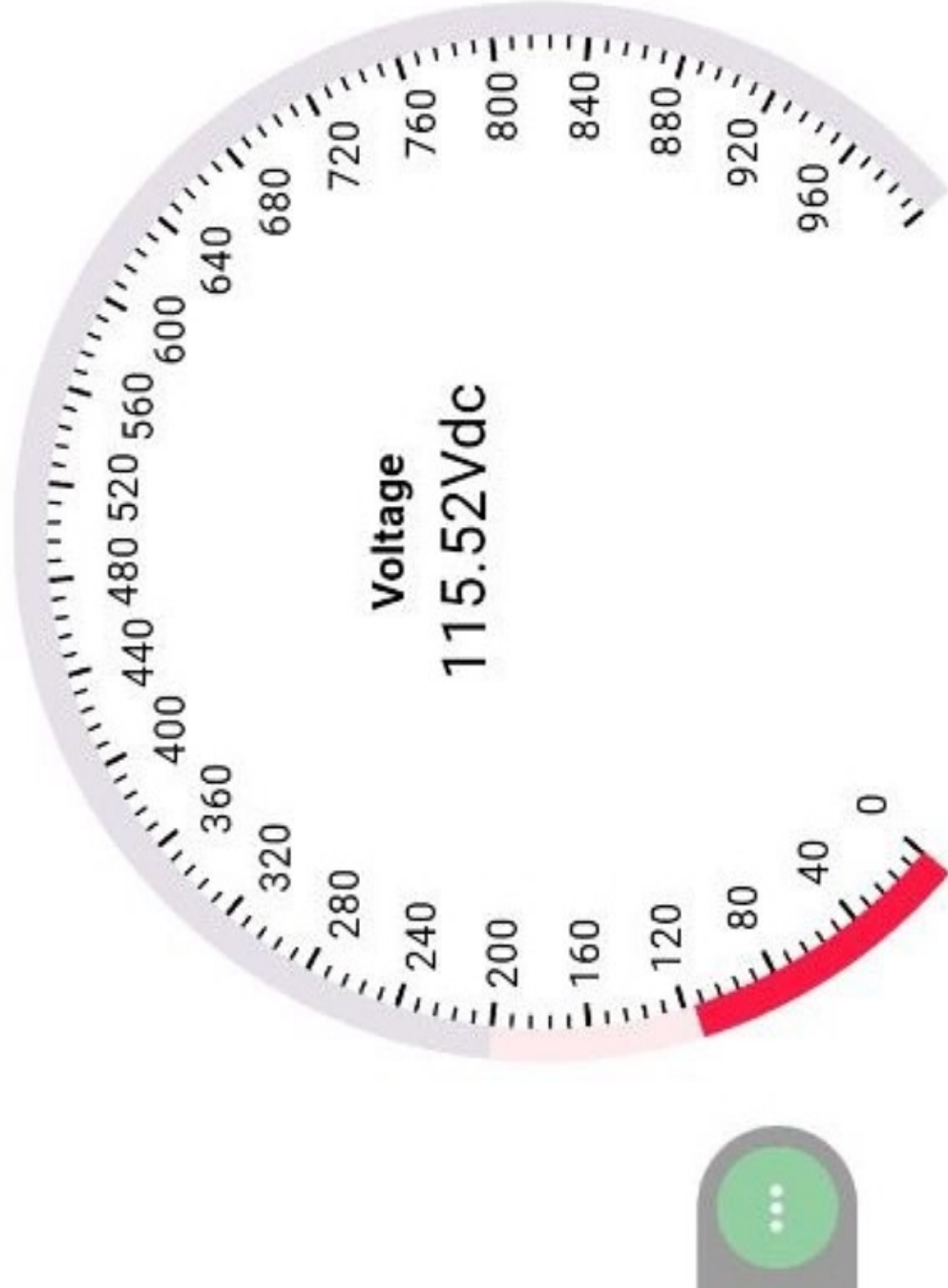
Resistance

10.5,40,30

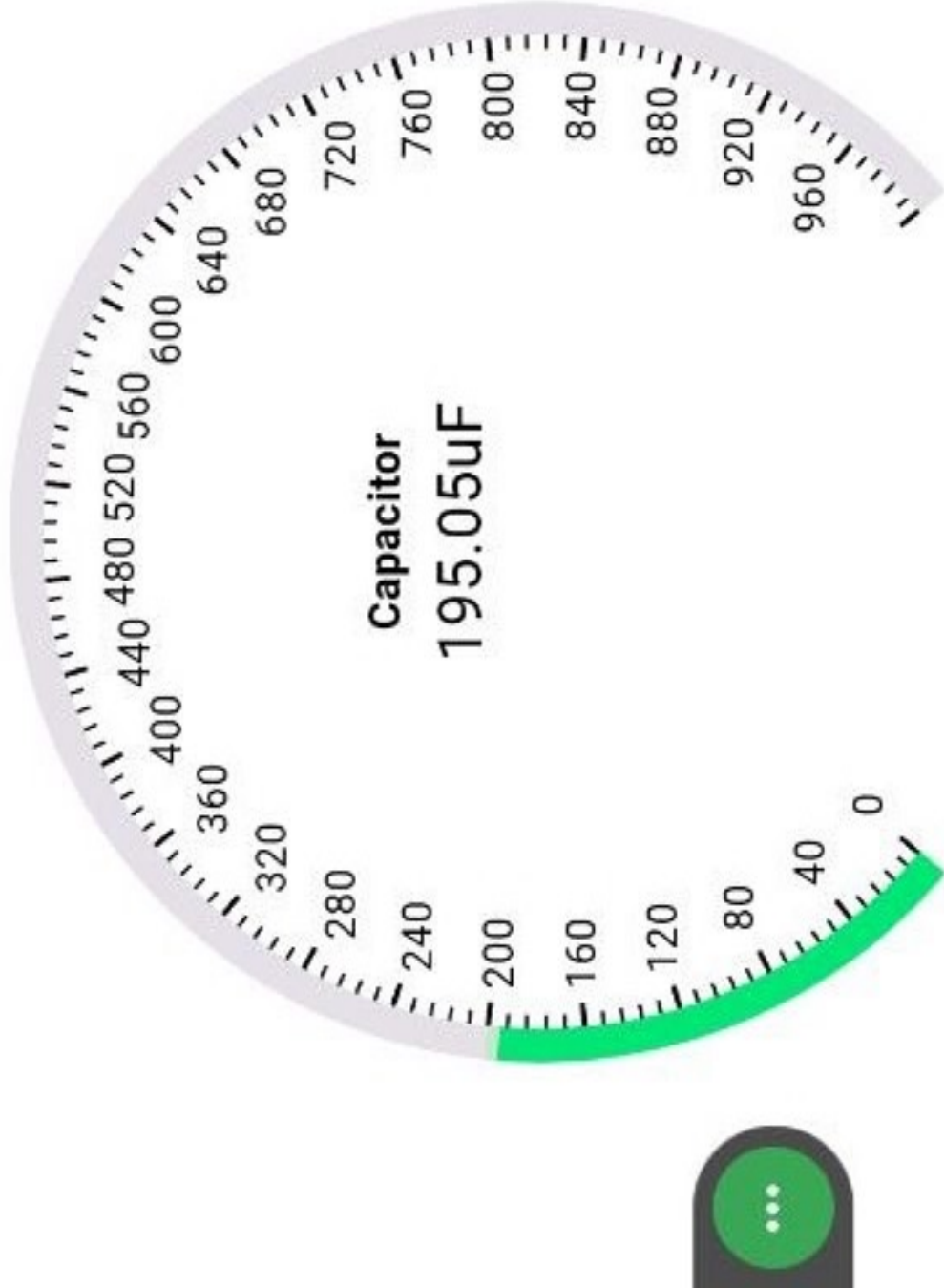
CALCULATE

Result: **35.30Ω**

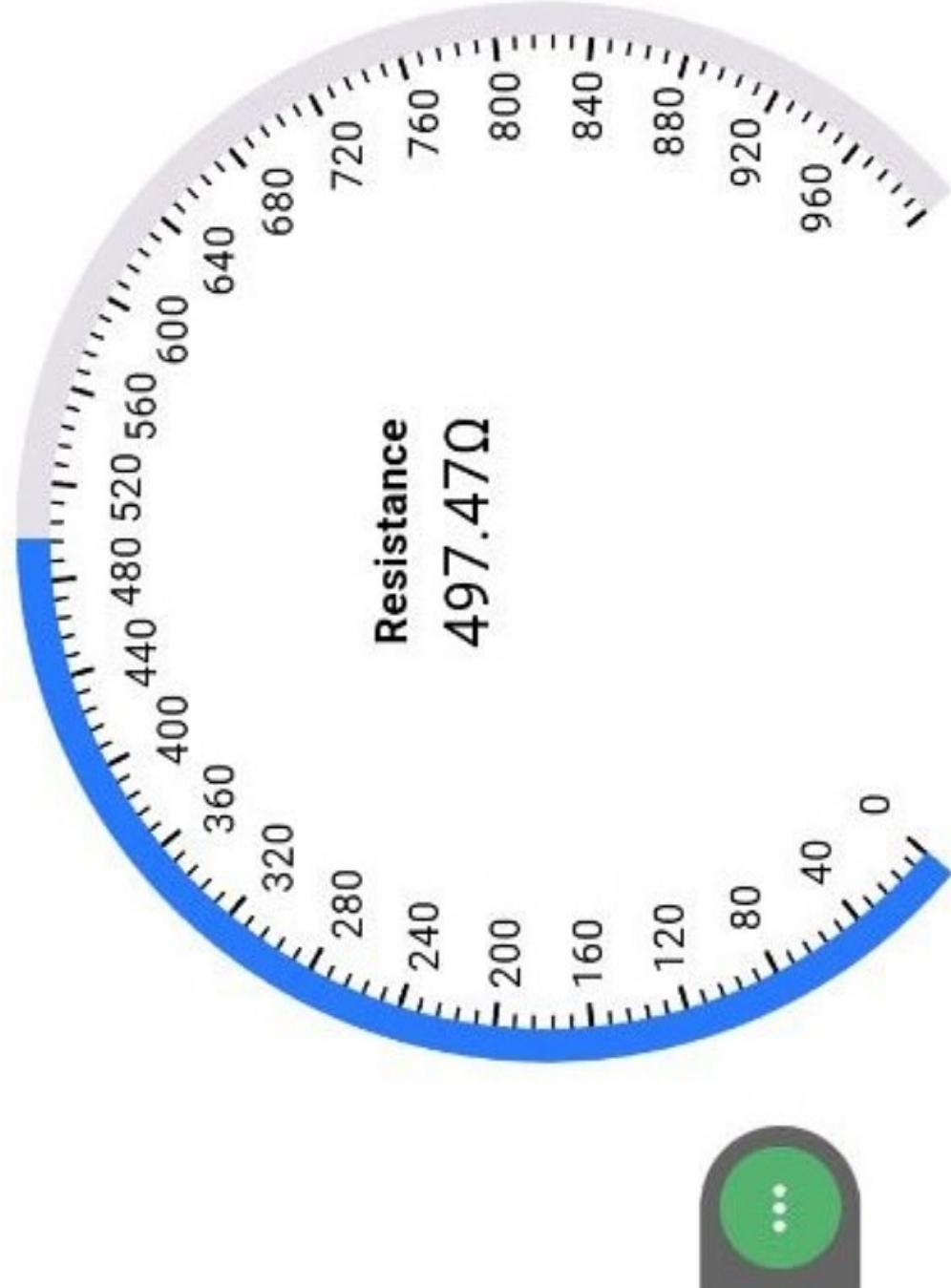




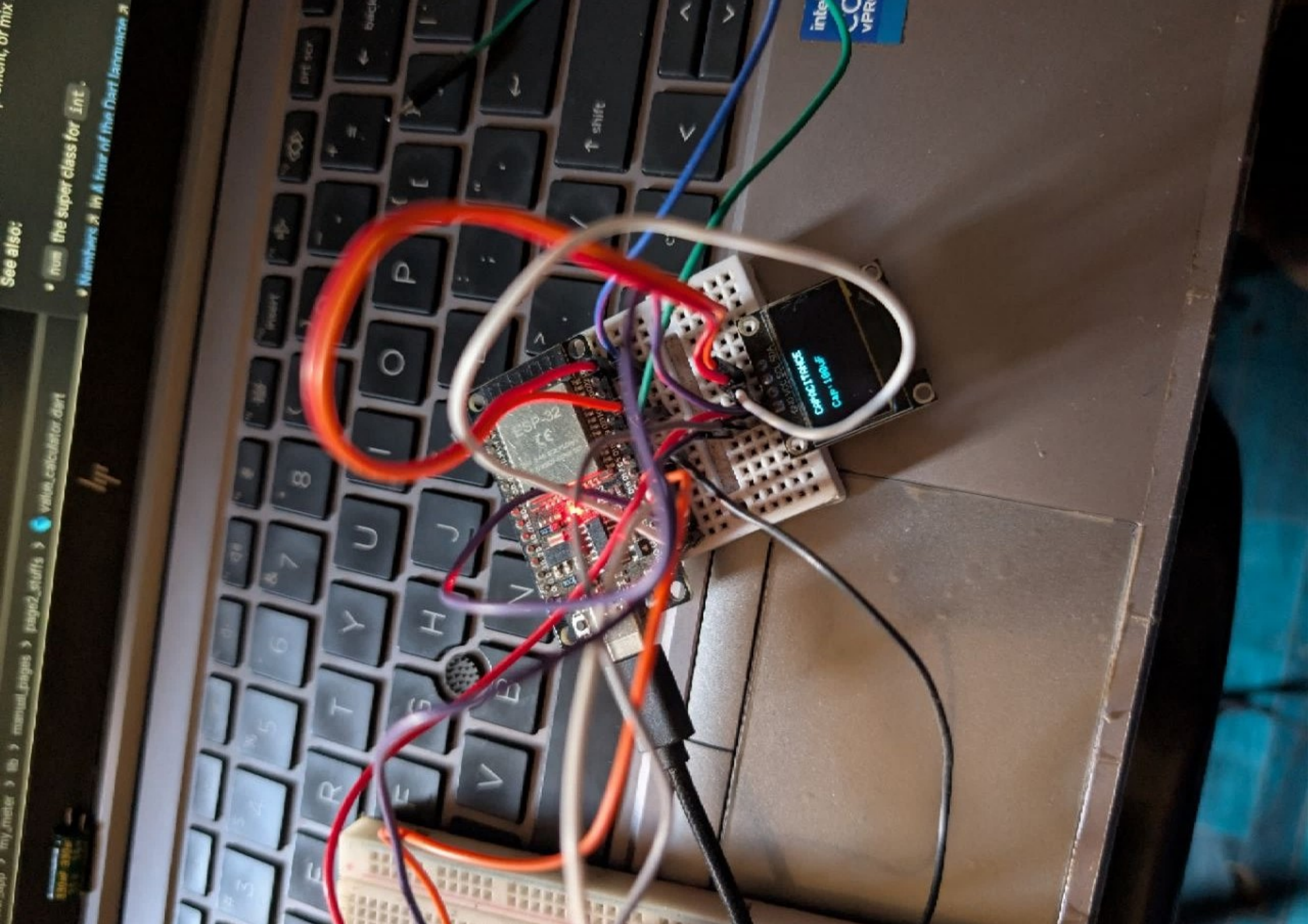
Result: 115.52Vdc



Result: **195.05uF**



Result: 497.47Ω



See also:

- `num` the super class for `int`.
- `Numbers` in A tour of the Dart language

value, calculator.dart

330w app

```

}

float avgAdc(){//helps take average samples reading
    float rawReading=0.00;
    for(int x=0; x<16; x++){//get average adc value after taking numbers of cycles
        rawReading+=analogRead(ohmAdc);
    }
    return rawReading/16.0f;
}

float map_it(float val, float in_min, float in_max, float out_min, float out_max){//mapping method for floating point values
    float result=(val-in_min)*(out_max-out_min)/(in_max-in_min)+out_min;
    return result;
}

String unitOhm(const float value){//method helps return ohm and unit
    if(value <1000) return "ohm";
    else if(value>1000 && value<1000000) return "Kohm";
    else {return "Mohm";}
}

void getResistance(bool isContinuity , bool isDiode){//method helps get and set diode, continuity and resistance values
    float adcAvg = avgAdc();//get average ADC reading
    if(adcAvg<1020){
        float v_out= map_it(adcAvg, 0.00, 1023.00, 0.00, v_in);
        float R2_value=(Rtop*v_out)/(v_in-v_out);
        if(isContinuity && R2_value< 40.0f) digitalWrite(BUZZ_PIN, HIGH);//sound alarm buzzer if continuity if below 40 ohm
        else{digitalWrite(BUZZ_PIN, LOW);}
        String unit = unitOhm(R2_value);
        float resistance= unit=="ohm"? R2_value : unit=="Kohm"? R2_value/1000.0f : R2_value/1000000.0f ;
        String result=isDiode? String(v_out)+"v": String(resistance)+unit;
        if(millis()-newTime>=interval){
            newTime=millis();
            isDiode? Serial.println("Diode drop:"+String(result)):Serial.println("R2:"+String(R2_value));
        }
    }
    else if(isContinuity) digitalWrite(BUZZ_PIN, LOW);
}

```

```
45     break;
46     case 4: onOff(0);
47     break;
48     default: onOff(0);
49 }
50 }
51
52 void operation(uint8_t value ){ //helps select suitable method based off selected page function
53     switch(value){
54         case 0: myMeter.getResistance(false, false, setValues);
55         break;
56         case 1: myMeter.getCapacitance(setValues);
57         break;
58         case 2: myMeter.getVolt(seValues);
59         break;
60         case 3: myMeter.getResistance(true, false, setValues);
61         break;
62         case 4: myMeter.getResistance(false, true, setValues);
63     }
64 }
65
66 void loop() {
67     myMeter.getDisplay(&comNum, &pageNum, &valueData, &unitData);
68     chooseMode(conNum);
69     if(pageNum==3){//if we are ready to send data
70         operation(comNum);
71         myBle.pushData(valueData, unitData, 2000);
72     }
73     if(startBle){
74         myBle.startServiceAds();
75         startBle=false;
76     }
77     // put your main code here, to run repeatedly:
78 }
```